

Shopfast E-Commerce Platform — Kubernetes Implementation Report

Date: 2026-08-17

Cluster: Single-node k3s on 72.62.250.5 (srv1286917)

GitOps: ArgoCD — <https://github.com/JunaidIshaq/ecommerce.git> → k8s/overlays/dev

Ingress: nginx + self-signed cert-manager TLS

Platform: Spring Cloud microservices + Angular frontend + Keycloak OIDC

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1. Cluster Architecture

1.1 Infrastructure

Layer	Component	Details
Kubernetes distro	k3s (v1.x)	Single-node, containerd runtime
Node	srv1286917 (72.62.250.5)	4 CPU, single node
Container registry	Docker Registry	localhost:5000 (insecure, node-local)
Ingress controller	ingress-nginx	Exposed via host network on 80/443
TLS	cert-manager + self-signed	Issuer: shopfast-selfsigned , Cert: shopfast-tls
GitOps	ArgoCD	3 applications managing the cluster

1.2 ArgoCD Applications

Application	Source Path	Sync Status	Health
shopfast-services	k8s/overlays/dev	OutOfSync	Degraded (shared-resource warnings)
shopfast-infra	k8s/overlays/dev	OutOfSync	Degraded (shared-resource warnings)
shopfast-ingress	k8s/ingress	Synced	Healthy

Note: "Degraded" is caused by shared resources (Ingress, Certificate, NetworkPolicies, Namespace) declared across multiple ArgoCD apps. Pods themselves are healthy.

1.3 Tech Stack

Layer	Technology
Frontend	Angular 19 + angular-auth-oidc-client v19

Layer	Technology
API Gateway	Spring Cloud Gateway
Service Discovery	Netflix Eureka Server
Business Services	Spring Boot 3.x (14 services)
Identity	Keycloak 26 (OIDC / OAuth2)
Message Queue	Kafka 7.6.1 + ZooKeeper
Search	Elasticsearch 8.19.5
Cache	Redis 7
Databases	PostgreSQL 15
Monitoring	(planned) Prometheus + Grafana via HPA

2. Namespace & Workloads

All application workloads run in the `shopfast` namespace.

2.1 Deployments (Complete List)

Deployment	Image	Replicas	Status	Restarts
frontend	localhost:5000/shopfast/frontend:dev	1	Running	0
api-gateway	localhost:5000/shopfast/api-gateway:2d9e2e2	1	Running	0
keycloak	quay.io/keycloak/keycloak:26.0	1	Running	0
eureka-server	localhost:5000/shopfast/eureka-server:dev	1	Running	0
product-service	localhost:5000/shopfast/product-service:dev	1	Running	3
cart-service	localhost:5000/shopfast/cart-service:dev	1	Running	0
order-service	localhost:5000/shopfast/order-service:dev	1	Running	2
user-service		1	Running	0

Deployment	Image	Replicas	Status	Restarts
	localhost:5000/shopfast/ user-service:dev			
auth-service	localhost:5000/shopfast/ auth-service:dev	1	CrashLoopBackOff	14
admin-service	localhost:5000/shopfast/ admin-service:dev	1	Running	0
category-service	localhost:5000/shopfast/ category-service:dev	1	Running	0
inventory-service	localhost:5000/shopfast/ inventory-service:dev	1	Running	0
payment-service	localhost:5000/shopfast/ payment-service:dev	1	Running	0
coupon-service	localhost:5000/shopfast/ coupon-service:dev	1	Running	0
review-service	localhost:5000/shopfast/ review-service:dev	1	Running	0
notification-service	localhost:5000/shopfast/ notification-service:dev	1	Running	0
elastic-service	localhost:5000/shopfast/ elastic-service:dev	1	Running	0
postgres	postgres:15	1	Running	0
redis	redis:7	1	Running	0
kafka	confluentinc/cp-kafka:7.6.1	1	Running	68
zookeeper	confluentinc/cp-zookeeper: 7.6.1	1	Running	0
elasticsearch	docker.elastic.co/ elasticsearch:8.19.5	1	Running	0
pgadmin	dpage/pgadmin4:latest	1	Running	0
kafka-ui	provectuslabs/kafka- ui:latest	1	Running	0

Note: Kafka's 68 restarts indicate instability — likely resource pressure or ZooKeeper connectivity issues.

2.2 Deployment Strategy

- **Replicas:** All services run with 1 replica (single-node cluster constraint).
- **Image Pull Policy:** `IfNotPresent` (uses local registry).
- **Build Context:** Docker multi-stage builds on the server (`/root/ecommerce/`), pushed to `localhost:5000` .
- **Frontend:** Angular production build → `nginx:alpine`.

3. Services & Networking

3.1 Service Registry

All internal services use ClusterIP services with Eureka for service discovery:

Service	ClusterIP	Port(s)	Protocol
api-gateway	10.43.33.121	8080	TCP
frontend	10.43.254.130	80	TCP
keycloak	10.43.223.111	8180, 9000	TCP
eureka-server	10.43.160.59	8761	TCP
product-service	10.43.79.72	8081	TCP
cart-service	10.43.95.242	8088	TCP
order-service	10.43.237.47	8084	TCP
auth-service	10.43.161.230	8087	TCP
user-service	10.43.146.197	8086	TCP
category-service	10.43.235.67	8082	TCP
inventory-service	10.43.102.46	8083	TCP
payment-service	10.43.243.44	8085	TCP
coupon-service	10.43.57.31	8089	TCP
review-service	10.43.49.115	8090	TCP
notification-service	10.43.78.144	8091	TCP
elastic-service	10.43.17.93	8092	TCP
admin-service	10.43.125.64	8093	TCP

Service	ClusterIP	Port(s)	Protocol
postgres	10.43.56.199	5432	TCP
redis	10.43.217.166	6379	TCP
kafka	10.43.123.69	9092	TCP
elasticsearch	10.43.108.234	9200, 9300	TCP
pgadmin	10.43.233.81	80	TCP
kafka-ui	10.43.81.158	8080	TCP

3.2 Service Communication

- **Frontend → API:** HTTPS via ingress (/api → api-gateway:8080)
- **Gateway → Services:** HTTP via Eureka discovery or direct service DNS
- **Service-to-Service:** Feign clients with `ServiceTokenRelayInterceptor` (OAuth2 client_credentials)
- **Config:** `shopfast-config` ConfigMap injected as env vars; secrets via `shopfast-secrets`

4. Ingress & TLS

4.1 Ingress Controller

- **Class:** `nginx`
- **Address:** `72.62.250.5`
- **Ports:** 80 (HTTP), 443 (HTTPS)
- **TLS:** Self-signed certificate (`shopfast-tls` secret)

4.2 Routing Rules

Host	Path	Backend Service	Port
shopfast.live	/api	api-gateway	8080
shopfast.live	/realms	api-gateway → Keycloak proxy	8080
shopfast.live	/	frontend	80
shopfast.live	/eureka	eureka-server	8761
shopfast.live	/kafka	kafka-ui	8080

Host	Path	Backend Service	Port
shopfast.live	/pgadmin	pgadmin	80
www.shopfast.live	/	frontend	80
shopfast-test.live	/api , /realms , /	api-gateway / frontend	8080/80
www.shopfast-test.live	/	frontend	80

4.3 TLS Configuration

```
tls:
  - hosts:
    - shopfast.live
    - www.shopfast.live
    secretName: shopfast-tls
```

- **Issuer:** shopfast-selfsigned (self-signed ClusterIssuer)
- **Certificate:** shopfast-tls (managed by cert-manager)
- **Production TODO:** Replace with Let's Encrypt ClusterIssuer

4.4 Frontend Nginx Config

The frontend-nginx-config ConfigMap provides: - SPA fallback routing (all non-API requests → index.html) - Legacy proxy paths (/api , /users , /orders → admin-service:8093)

Issue: The /api path in frontend-nginx bypasses the api-gateway for admin paths. This is a legacy configuration that should be reviewed.

5. Configuration & Secrets

5.1 ConfigMaps

ConfigMap	Keys	Purpose
shopfast-config	26 keys	DB, Redis, Kafka, Keycloak (issuer/base/jwk), Eureka URLs
keycloak-realm	1 key (realm-export.json)	Empty (0 bytes) — realm import may be broken
	1 key	Elasticsearch URI

ConfigMap	Keys	Purpose
elasticsearch-config		
frontend-nginx-config	1 key (default.conf)	Nginx SPA routing
postgres-init	SQL scripts	Database initialization

5.2 ConfigMap Key Values (shopfast-config)

```

KEYCLOAK_ISSUER_URI    = https://shopfast.live/realms/shopfast
KEYCLOAK_BASE_URL      = http://keycloak:8180
KEYCLOAK_JWK_SET_URI   = http://keycloak:8180/realms/shopfast/protocol/openid-connect/certs
KEYCLOAK_REALM         = shopfast
KEYCLOAK_ADMIN_CLIENT_ID = shopfast-services
POSTGRES_HOST          = postgres
POSTGRES_PORT          = 5432
REDIS_HOST              = redis
REDIS_PORT              = 6379
KAFKA_BOOTSTRAP        = kafka:9092
EUREKA_URL              = http://eureka-server:8761/eureka/eureka/

```

5.3 Secrets (shopfast-secrets)

Key	Purpose
DB_PASSWORD	PostgreSQL password
JWT_SECRET	Symmetric JWT fallback/legacy
KEYCLOAK_ADMIN	Keycloak admin username
KEYCLOAK_ADMIN_PASSWORD	Keycloak admin password
SHOPFAST_SERVICES_CLIENT_SECRET	Service-to-service client_credentials
SHOPFAST_ADMIN_CLIENT_SECRET	Admin client secret
APP_PASSWORD_ENCRYPTION_KEY	Application-level encryption key

6. Security

6.1 Identity & Access Management

Component	Details
IdP	Keycloak 26 (RS256)
Web client	shopfast-web (public, PKCE)
Service client	shopfast-services (confidential, client_credentials, serviceAccountsEnabled=true)
Admin client	shopfast-admin (confidential)
Token issuer	https://shopfast.live/realms/shopfast
Audience mapper	shopfast-web has audience mapper → aud: ["shopfast-web", "account"]

6.2 Frontend Authentication

- **Library:** angular-auth-oidc-client v19
- **Flow:** Authorization Code + PKCE
- **Token storage:** SessionStorage (not localStorage)
- **Silent renew:** Enabled (120s before expiry)
- **Custom interceptor:** tokenInterceptor attaches bearer token to /api/** requests synchronously

6.3 Backend Resource Servers

- All services validate JWT against https://shopfast.live/realms/shopfast
- /api/v1/product/internal/** endpoints require .authenticated() (service-to-service)
- Public endpoints (GET /api/v1/product/**) are permitAll
- Admin endpoints require ROLE_ADMIN

6.4 Service-to-Service Authentication

- **Mechanism:** ServiceTokenRelayInterceptor (common-lib)
- **Client:** shopfast-services (client_credentials grant)
- **Fallback:** Relays caller's user token when present
- **Activation:** Conditional on shopfast.security.service-client.client-secret being set

6.5 Network Policies

Policy	Purpose
default-deny-ingress	Deny all ingress by default
allow-ingress-to-edge	Allow ingress-nginx to reach api-gateway, frontend, eureka, kafka-ui, pgadmin
allow-intra-namespace	Allow all pods to communicate within namespace (Eureka/Feign/Kafka)
allow-egress	Allow DNS + external egress

6.6 Security Gaps

Gap	Severity
No RBAC (ServiceAccounts, Roles, RoleBindings)	High
Self-signed TLS only (no Let's Encrypt)	Medium
Secrets in plain Kubernetes Secrets (not Sealed Secrets)	Medium
No PodSecurity Standards enforcement	Medium
frontend-nginx-config has stale /api paths bypassing gateway	Low
No image scanning in CI/CD	Low

7. Storage

7.1 Persistent Volume Claims

PVC	Size	Usage
keycloak-data	4Gi	Keycloak realm data + user sessions
(postgres PVC)	(not verified)	Cart, Order, User databases

7.2 Storage Classes

- Not explicitly defined — uses k3s default storage class.
- No backup/restore strategy configured.

8. Autoscaling

8.1 Horizontal Pod Autoscalers

HPA	Target	Min	Max	Current
api-gateway-hpa	CPU 70%	1	6	1
order-service-hpa	CPU 70%	1	6	1
product-service-hpa	CPU 70%	1	6	1

Note: HPA requires metrics-server. On single-node k3s, this is typically available by default.

9. GitOps (ArgoCD)

9.1 Applications

App	Repository	Path	Sync Policy
shopfast-services	https://github.com/JunaidIshaq/ecommerce.git	k8s/overlays/dev	Automated (Prune)
shopfast-infra	https://github.com/JunaidIshaq/ecommerce.git	k8s/overlays/dev	Automated (Prune)
shopfast-ingress	https://github.com/JunaidIshaq/ecommerce.git	k8s/ingress	Automated (Prune)

9.2 Kustomize Structure

```
k8s/
├── base/
│   ├── namespace.yaml
│   ├── configmap.yaml      # shopfast-config (shared env)
│   ├── network-policies.yaml
│   ├── quota.yaml
│   └── secrets/
│       └── secrets.yaml    # shopfast-secrets
├── services/               # 18 service manifests
│   ├── frontend.yaml
│   └── api-gateway.yaml
```

```
|   ├── keycloak (in infra)
|   └── ...
|   └── infra/
|       ├── postgres.yaml
|       ├── redis.yaml
|       ├── kafka.yaml
|       ├── keycloak.yaml
|       └── ...
|   └── ingress/
|       └── ingress.yaml
|   └── overlays/dev/
|       └── kustomization.yaml      # Retags ghcr.io → localhost:5000, scales to 1 replica
```

9.3 Image Tag Strategy (Dev)

Original Image	Dev Image
ghcr.io/shopfast/frontend	localhost:5000/shopfast/frontend:dev
ghcr.io/shopfast/api-gateway	localhost:5000/shopfast/api-gateway:2d9e2e2
ghcr.io/shopfast/eureka-server	localhost:5000/shopfast/eureka-server:dev
All other services	localhost:5000/shopfast/<service>:dev

Note: The api-gateway uses a pinned SHA (2d9e2e2) instead of dev tag.

9.4 Current Git State

- **HEAD:** f48826b — fix: AuthService.getAccessToken returns string
- **Unsynced commits:** 773026f (issuer alignment + frontend interceptor), f48826b (auth fix)
- **Sync Status:** OutOfSync from HEAD

10. Issues Fixed

Date	Issue	Root Cause	Fix Applied
2026-08-16	Keycloak login → callback loop	CallbackComponent called authorize() on checkAuth() error	Navigate to / on error; consume return URL
2026-08-16			

Date	Issue	Root Cause	Fix Applied
	userinfo 401 "iss claim is not valid"	KC_HOSTNAME unset → inconsistent issuer derivation	Set KC_HOSTNAME=https://shopfast.live
2026-08-16	userinfo 401 "audience"	No audience mapper on shopfast-web client	Added oidc-audience-mapper protocol mapper
2026-08-16	API Gateway crash (JWT validation)	Gateway KEYCLOAK_ISSUER_URI=http://keycloak:8180 vs Keycloak public issuer https://shopfast.live	Updated gateway env to public issuer
2026-08-16	Cart 500 / Service 401	Backend services validated tokens against internal issuer while Keycloak issued public-iss tokens	Updated KEYCLOAK_ISSUER_URI to https://shopfast.live in ConfigMap + all service manifests
2026-08-17	Frontend Bearer [object Object]	AuthService.getAccessToken() returned Observable cast to string	Read token synchronously from sessionStorage
2026-08-17	Frontend no token in API headers	Library interceptor not reliably attaching token	Added explicit tokenInterceptor

11. Remaining Gaps

11.1 Critical / Immediate

#	Gap	Impact	Recommendation
1	ArgoCD OutOfSync	Latest config/issuer changes not fully propagated	Trigger manual sync for shopfast-services and shopfast-infra
2	auth-service CrashLoopBackOff	Auth endpoints unstable	Investigate pod logs; old pod still serving
3	keycloak-realm ConfigMap empty	Realm may not auto-import from realm-export.json	Verify realm import config; fix ConfigMap
4			Review and update proxy targets

#	Gap	Impact	Recommendation
	frontend-nginx-config stale paths	<code>/api</code> , <code>/users</code> , <code>/orders</code> bypass api-gateway auth	
5	Kafka 68 restarts	Message queue instability	Check ZooKeeper connectivity, resource limits, logs

11.2 Security Hardening

#	Gap	Recommendation
1	No RBAC (ServiceAccounts, Roles, RoleBindings)	Implement least-privilege RBAC for all service accounts
2	Self-signed TLS only	Replace cert-manager Issuer with Let's Encrypt ClusterIssuer
3	Secrets in plain Kubernetes Secrets	Consider Sealed Secrets or external vault (HashiCorp Vault)
4	No PodSecurity Standards	Add <code>runAsNonRoot</code> , <code>readOnlyRootFilesystem</code> , <code>allowPrivilegeEscalation: false</code>
5	No image scanning	Add Trivy/Clair to CI pipeline
6	No network segmentation beyond 3 policies	Fine-trace NetworkPolicies per service

11.3 Observability & Operations

#	Gap	Recommendation
1	No Prometheus/Grafana	Add monitoring stack; HPA needs metrics-server
2	No centralized logging	Add ELK/Loki/PLG stack
3	No distributed tracing	Add OpenTelemetry instrumentation
4	No backup strategy	PVC snapshots for Postgres, Keycloak, Redis
5	No resource limits	Add <code>resources.requests/limits</code> to all deployments
6	No PodDisruptionBudgets	Ensure availability during node maintenance
7	No health check refinement	Add custom liveness/readiness probes beyond <code>/actuator/health</code>

11.4 CI/CD

#	Gap	Recommendation
1	No automated image build on push	Add GitHub Actions workflow
2	No automated ArgoCD sync	Add webhook or sync automation on merge
3	Image tags are <code>dev</code> /manual SHA	Use Git SHA or semantic version tags
4	No test stage	Add unit/integration tests before image build
5	No staging environment	Add <code>overlays/staging</code> for pre-prod validation

12. Current State Summary

```
Cluster:      UP (23/24 pods running, 1 CrashLoopBackOff)
Ingress:      UP (nginx + self-signed TLS)
Keycloak:     UP (public issuer https://shopfast.live/realms/shopfast)
API Gateway:  UP (public issuer validated)
Frontend:     UP (token interceptor deployed, auth fix deployed)
Auth Service: DEGRADED (CrashLoopBackOff, 14 restarts)
ArgoCD:       OutOfSync (manual sync needed)
Kafka:        UNSTABLE (68 restarts)
Eureka:       UP
Databases:    UP (Postgres, Redis)
```

Recommended Next Actions

1. **Trigger ArgoCD sync** for `shopfast-services` and `shopfast-infra` to apply latest issuer/config changes
2. **Investigate auth-service CrashLoopBackOff** — check logs for root cause
3. **Fix empty keycloak-realm ConfigMap** — ensure realm-export.json is loaded
4. **Clean up frontend-nginx-config** — remove stale `/api` , `/users` , `/orders` admin paths
5. **Investigate Kafka restarts** — check ZooKeeper connectivity and resource pressure
6. **Implement RBAC and Pod Security Standards**
7. **Replace self-signed TLS with Let's Encrypt**

Appendix A: Manifest Structure

```
k8s/
├── argocd/
│   ├── application-dev.yaml
│   ├── application-infra.yaml
│   ├── application-ingress.yaml
│   ├── application-services.yaml
│   └── applicationset.yaml
├── base/
│   ├── configmap.yaml           # shopfast-config (26 keys)
│   ├── kustomization.yaml
│   ├── namespace.yaml
│   ├── network-policies.yaml   # 4 policies
│   ├── quota.yaml
│   ├── sealed-secret-example.yaml
│   ├── servicemonitor.yaml
│   └── secrets/
│       ├── kustomization.yaml
│       └── secrets.yaml        # shopfast-secrets
├── infra/
│   ├── elasticsearch.yaml
│   ├── kafka.yaml
│   ├── keycloak-realm-configmap.yaml
│   ├── keycloak.yaml           # Keycloak 26 deployment
│   ├── kustomization.yaml
│   ├── pgadmin.yaml
│   ├── postgres.yaml
│   ├── realm-export.json
│   └── redis.yaml
├── ingress/
│   ├── ingress.yaml           # nginx ingress + cert-manager
│   └── kustomization.yaml
├── overlays/
│   ├── dev/
│   │   └── kustomization.yaml   # Dev overlay (retag images, scale 1)
│   └── prod/
│       ├── kustomization.yaml
│       └── prod-issuer.yaml
└── services/
    ├── admin-service.yaml
    ├── api-gateway.yaml
    ├── auth-service.yaml
    ├── cart-service.yaml
    ├── category-service.yaml
    ├── coupon-service.yaml
    ├── elastic-service.yaml
    ├── eureka-server.yaml
    ├── frontend-nginx-config.yaml
    ├── frontend.yaml
    ├── hpa.yaml                # HPA for gateway, order, product
    ├── inventory-service.yaml
    ├── kustomization.yaml
    ├── notification-service.yaml
    └── order-service.yaml
```

- └─ payment-service.yaml
- └─ product-service.yaml
- └─ review-service.yaml
- └─ user-service.yaml

Appendix B: Environment Variables (shopfast-config)

Variable	Value	Used By
KEYCLOAK_ISSUER_URI	https://shopfast.live/realms/shopfast	All services (JWT validation)
KEYCLOAK_BASE_URL	http://keycloak:8180	Admin/management calls
KEYCLOAK_JWK_SET_URI	http://keycloak:8180/realms/shopfast/protocol/openid-connect/certs	JWKS fetching
KEYCLOAK_REALM	shopfast	Keycloak realm name
POSTGRES_HOST	postgres	All DB services
POSTGRES_PORT	5432	All DB services
REDIS_HOST	redis	Cart, cache services
REDIS_PORT	6379	Cart, cache services
KAFKA_BOOTSTRAP	kafka:9092	Kafka producers/consumers
EUREKA_URL	http://eureka-server:8761/eureka/eureka/	All services

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